Happywhale - Whale and Dolphin Identification

Open-set individual identification with multi-view embedding retrieval

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Outline

- Introduction
- Related work
- Proposed Approach
- Experiment Result
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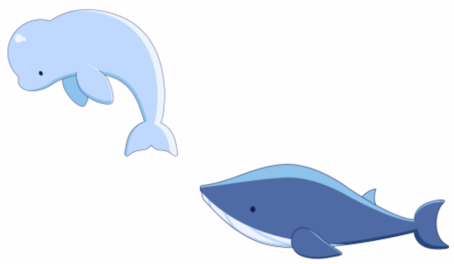
Whale and Dolphin Individual Identification

- **Goal:** Give an image of a whale or dolphin and predict the top-5 most likely individual identities.
- Critical to marine conservation, population monitoring, and large-scale photo-identification

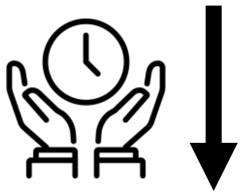
1. 60008f293a2b
2. bf2e0d964354 ✓
3. 9a66d006b36e
4. 641184a532a3
5. e8ac23241df5

Important

- **Why individual identification matters?**
  - Supports marine conservation and ecological research
  - Enables long-term tracking of whale and dolphin populations
  - Helps study migration, behavior, and population changes



- **Why automation is needed?**
  - Manual photo-identification requires expert knowledge
  - Large image databases make manual matching time-consuming
  - Automated retrieval can reduce human effort and scale up monitoring



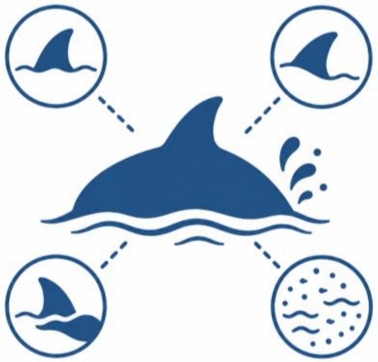
Challenges

Q1: Fine-grained cues

Similar appearances:
small scars, fin edges, and markings matter

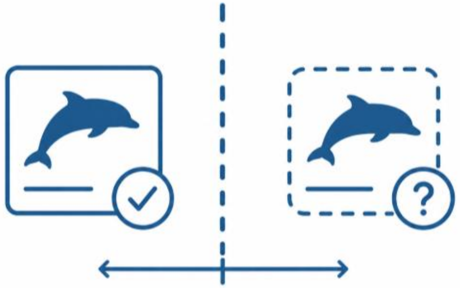
Q3: Long-tailed identities

Many identities, but most have only a few images



Q2: Real-world noise

Viewpoint, lighting, occlusion, water splash, and background noise



Q4: Open-set setting

Need to recognize known individuals and detect **new_individual**

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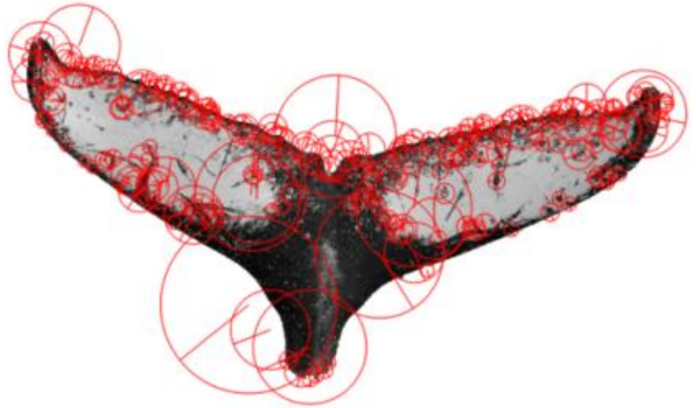
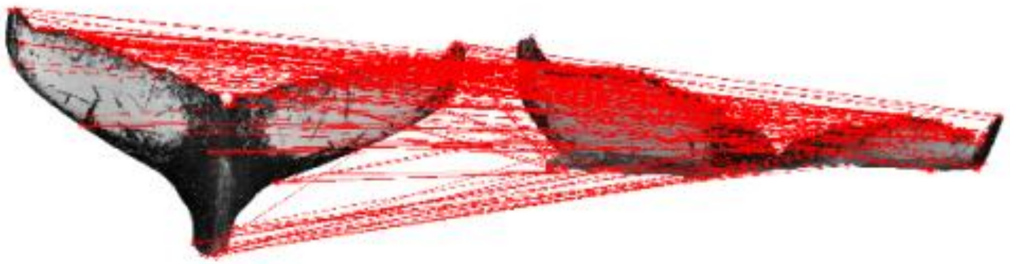
Traditional / Species-specific Photo-ID Methods

- **Local feature matching: SIFT+FLANN**
  - Detect distinctive keypoints from an image
  - Extract local descriptors around each keypoint
  - Match descriptors with images in the database
  - Predict identity based on reliable local matches



- **Pros:**
  - No large-scale training required
  - Interpretable matching process
  - Useful when fins, scars, or markings are clear

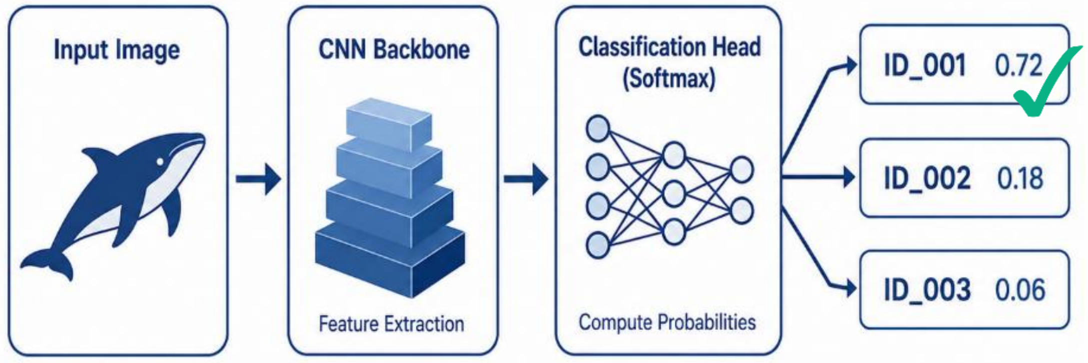


- **Cons:**
  - Sensitive to viewpoint and image quality
  - Hard to scale to thousands of identities

Jaisakthi et al., "Automatic Whale Matching System using Feature Descriptor," CLEF Working Notes, 2017.

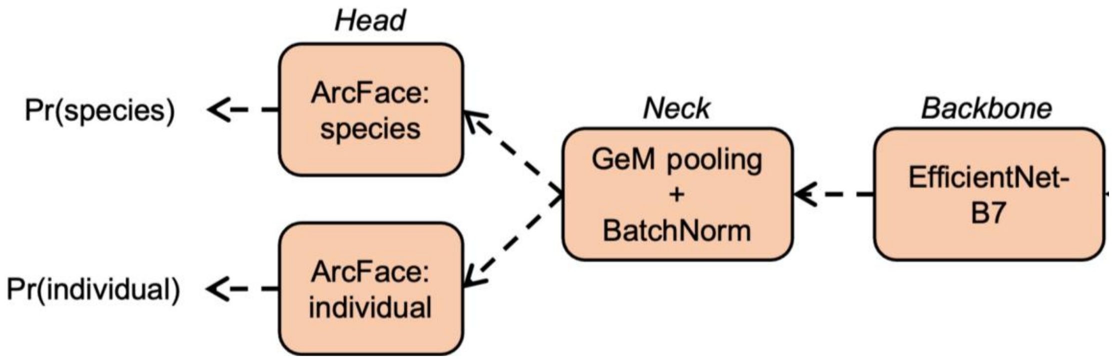
Closed-set Identity Classification

- **Description:**
  - Treat each individual as one class
  - Train a CNN classifier to predict individual_id
  - Output a probability distribution over known identities
- **Pipeline: CNN Backbone → Classification Head → individual_id**
- **Pros:**
  - Simple and direct training objective
  - Learns identity-level supervision
  - Works well when all test identities are known
- **Cons:**
  - Cannot naturally handle unseen individuals
  - Classifier is limited to training identities
  - Strongly affected by long-tailed identity distribution



Embedding-based Retrieval / Metric Learning

- **Description:**
  - Learn an embedding space for individual identification
  - Images of the same individual are close
  - Images of different individuals are separated
  - Use KNN retrieval during inference
- **Methods:**
  - ArcFace/Sub-center ArcFace
  - Adaptive margin



- **Pros:**
  - Suitable for open-set identification
  - Can retrieve similar known identities
  - Allows thresholding for new_individual
- **Cons:**
  - Needs careful training and score calibration
  - Sensitive to crop quality and embedding robustness

Deng et al., "ArcFace: Additive Angular Margin Loss for Deep Face Recognition," CVPR 2019.
An et al., "Sub-center ArcFace: Boosting Face Recognition by Large-Scale Noisy Web Faces," ECCV 2020.
Zhao, Zheng, et al. "An Individual Identification Method for Cetaceans Based on EfficientNet." 2024.

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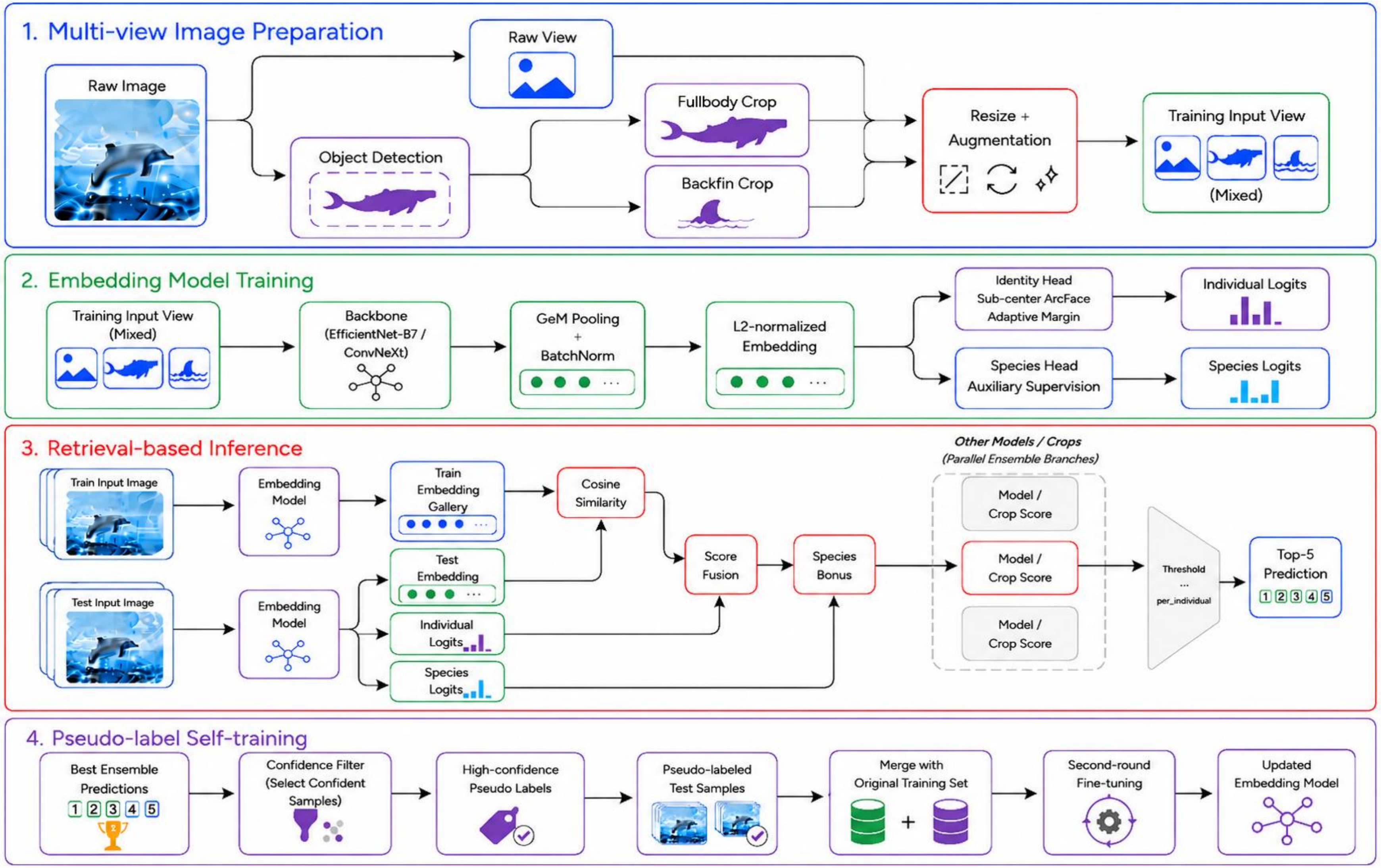
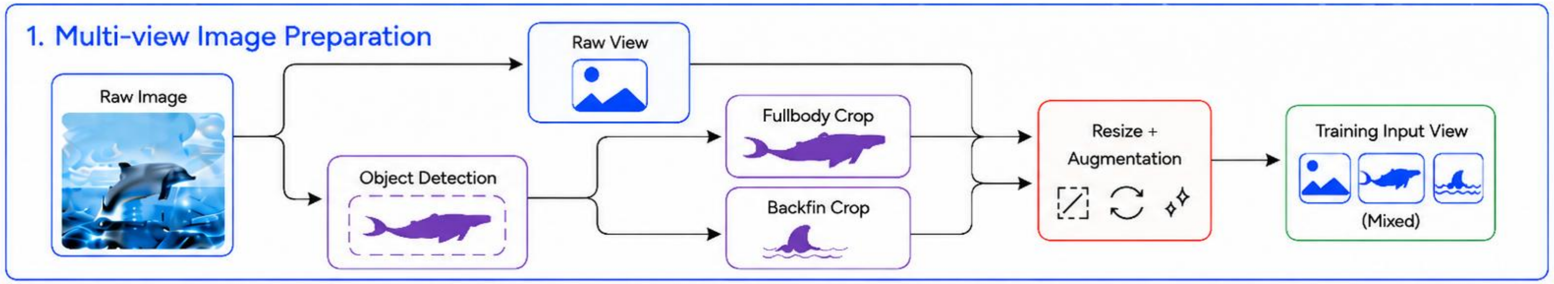
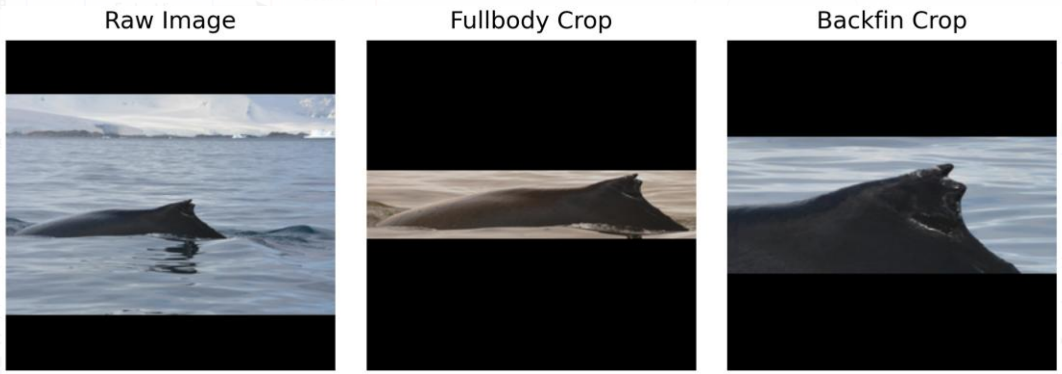
Approach Overview



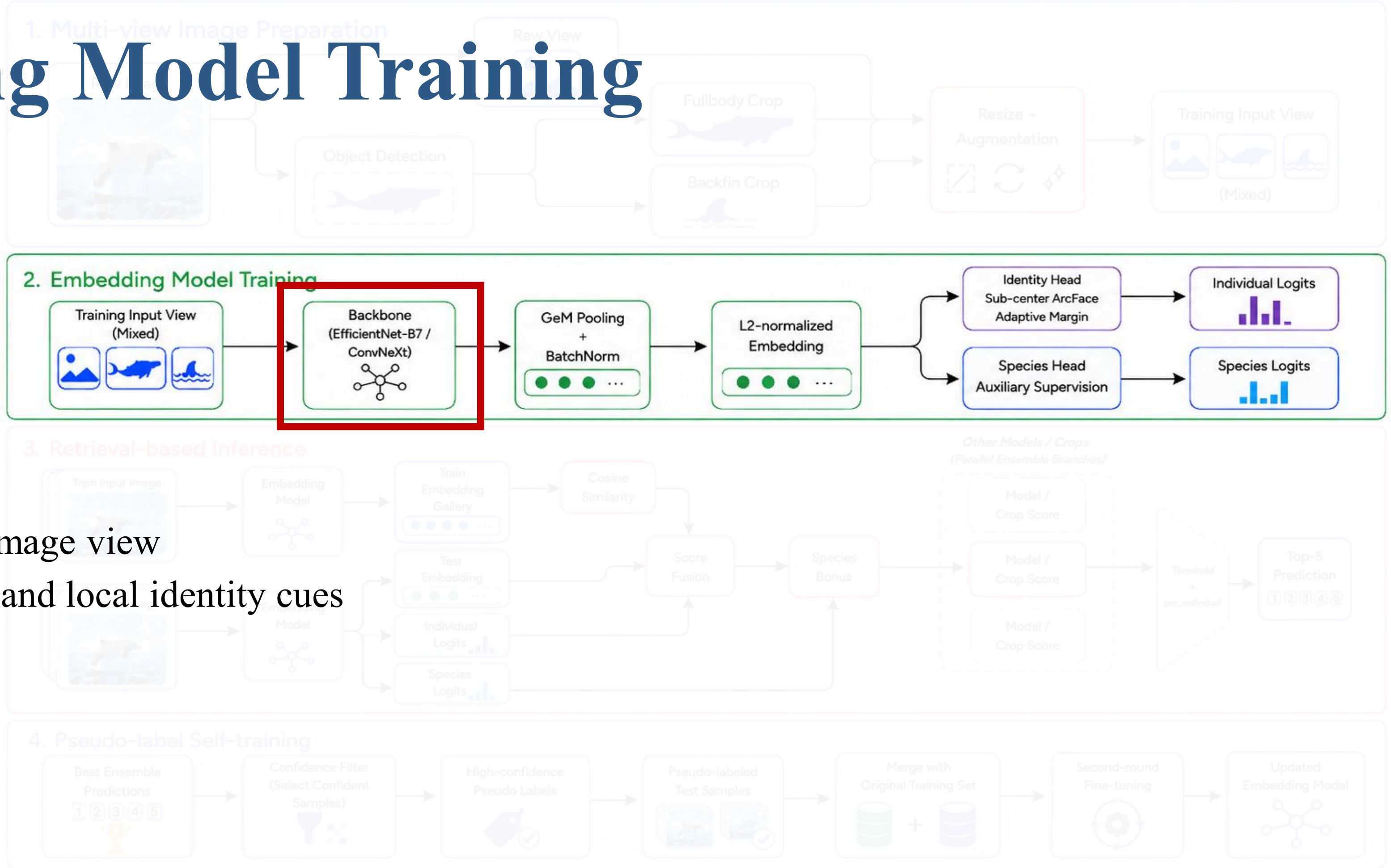
Stage 1: Multi-view Image Preparation



- **Motivation**
  - Identity cues appear in different body regions
  - Raw images contain context but also **noisy backgrounds**
  - Cropped views help focus on informative animal regions
- **Views**
  - Raw image: preserves global context
  - Fullbody crop: focuses on visible animal body
  - Backfin crop: emphasizes fin-related local cues
- **Implementation**
  - Use publicly available YOLOv5-based crop results
  - We do not train the detector ourselves
  - Crops are used as additional views for training and inference

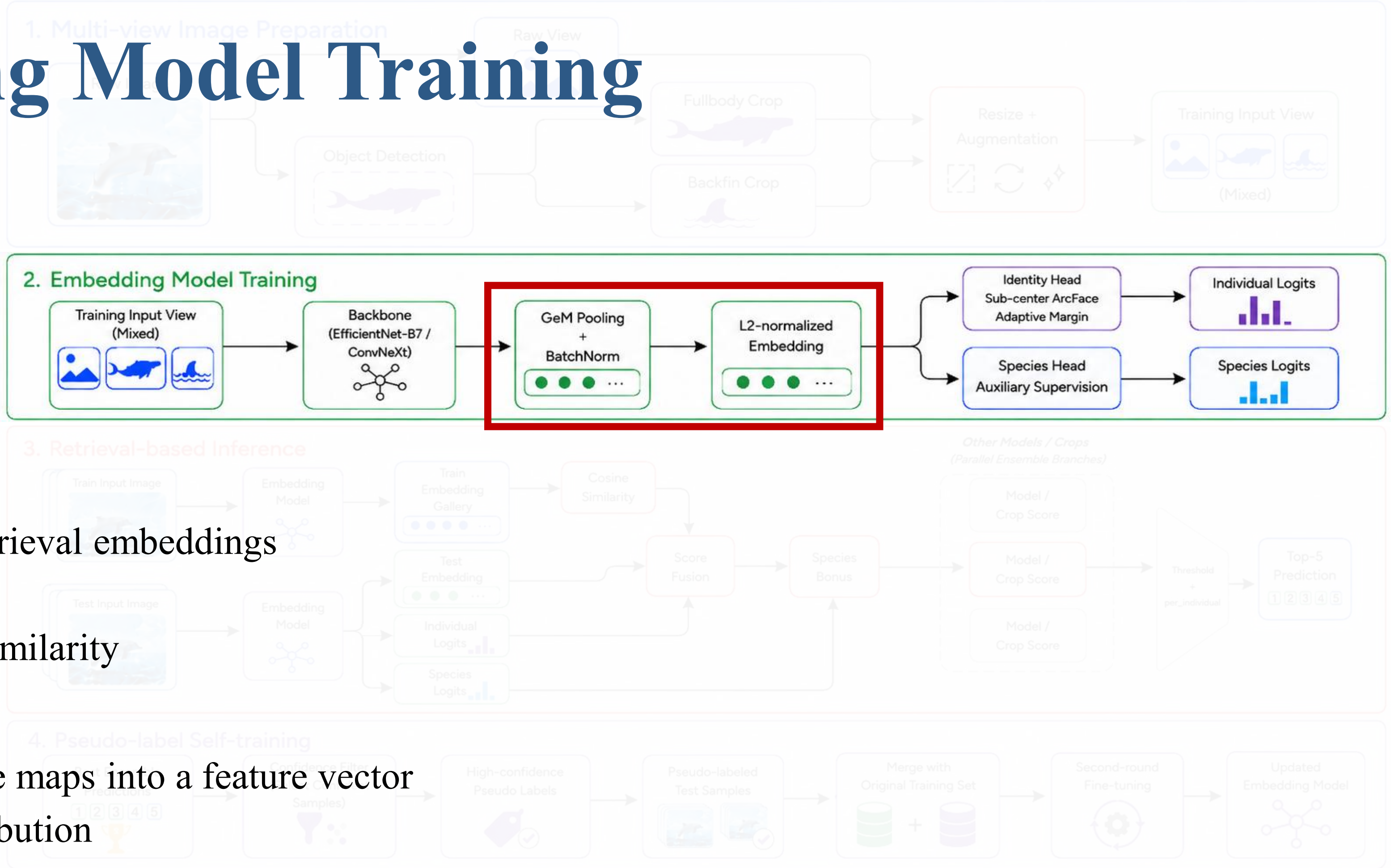


Stage 2: Embedding Model Training



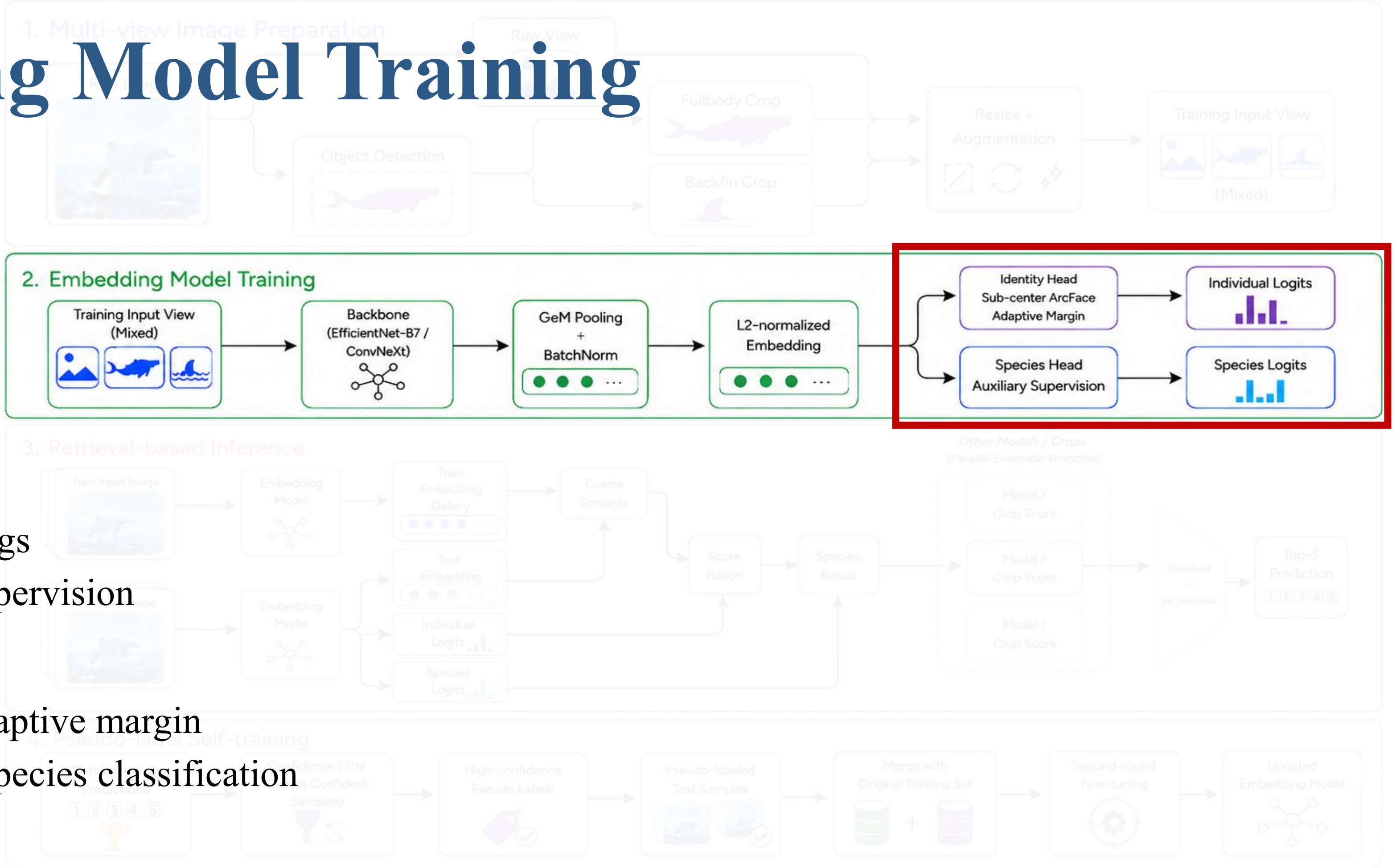
- **Backbone**
- **Goal:**
  - Extract visual feature maps from each image view
  - Capture both global animal appearance and local identity cues
- **Models:**
  - EfficientNet-B6
  - EfficientNet-B7
  - ConvNeXt-B
- **Output: Feature map from the pretrained backbone (C × H × W)**

Stage 2: Embedding Model Training



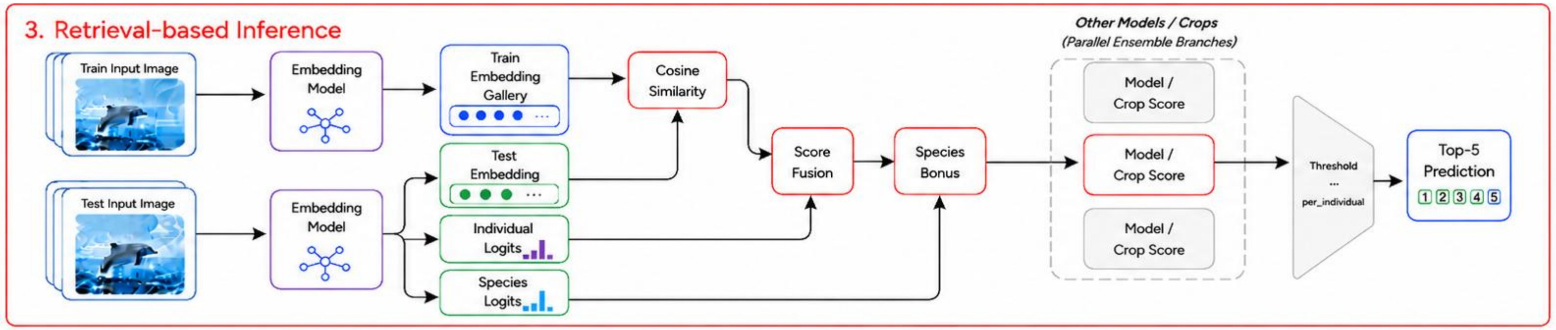
- **GeM + BN + L2 Normalization**
- **Goal:**
  - Convert backbone feature maps into retrieval embeddings
  - Emphasize discriminative local cues
  - Make embeddings suitable for cosine similarity
- **Implementation:**
  - GeM pooling: aggregates spatial feature maps into a feature vector
  - BatchNorm: stabilizes the feature distribution
  - L2 normalization: converts the vector into a unit-length embedding
- **Output: Normalized embedding vector (C)**

Stage 2: Embedding Model Training



- **Identity and Species Heads**
- **Goal:**
  - Learn discriminative identity embeddings
  - Use species information as auxiliary supervision
- **Implementation:**
  - Identity head: Sub-center ArcFace + adaptive margin
  - Species head: Sub-center ArcFace for species classification
- **Why Sub-center ArcFace?**
  - Multiple centers per identity
  - Handles viewpoint, crop, and pose variations
  - Separates different identities in angular space

Stage 3: Retrieval-based Inference



- **Goal:**
  - Generate top-5 predictions for each test image
  - Recognize known individuals and detect new_individual

1. KNN Retrieval

- Extract embeddings for train and test images
- Compare query embedding with the training gallery
- Use cosine similarity to retrieve nearest identities

2. Score Fusion and Ensemble

- Combine KNN similarity with identity logits
- Fuse different crop views: raw / fullbody / backfin
- Ensemble different models: B7 / B6 / ConvNeXt-B

$$S_{final} = \alpha S_{KNN} + (1 - \alpha) S_{logit}$$

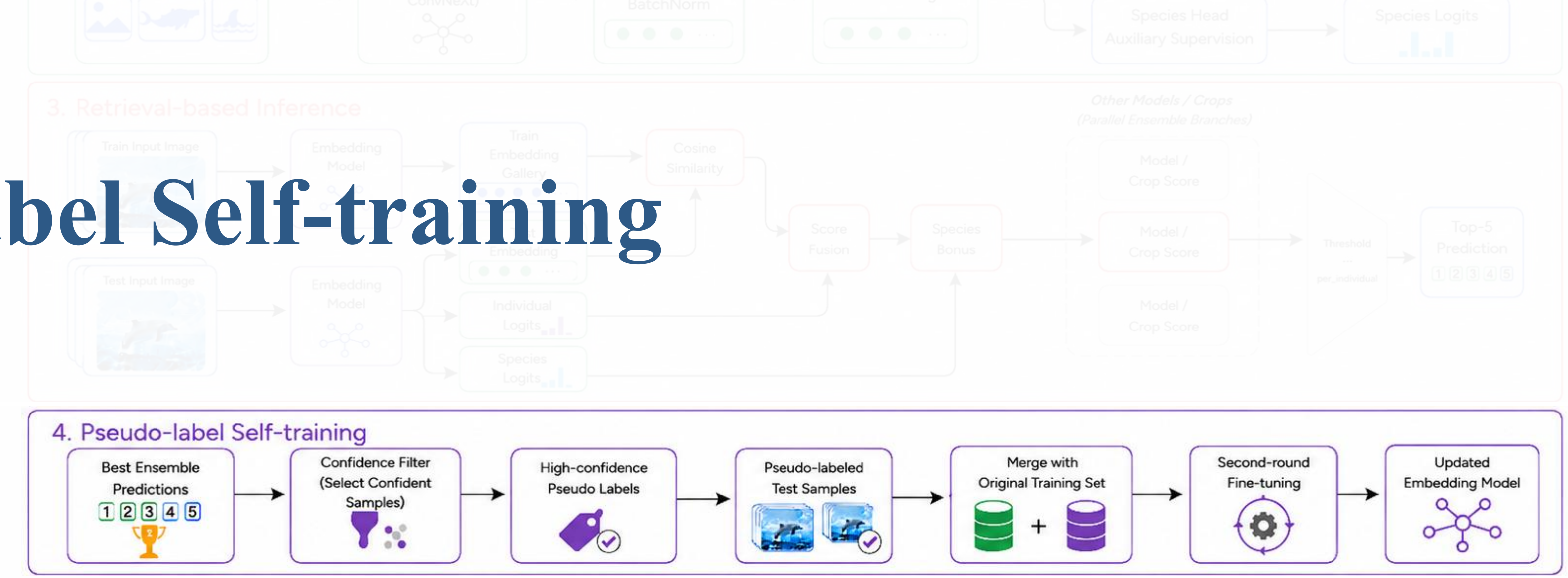
3. Species Bonus

- Rank known identities and new_individual together

4. new_individual Thresholding

- Treat new_individual as an extra candidate
- Add it with a threshold score
- Rank known identities and new_individual together

Stage 4: Pseudo-label Self-training



- **Goal:**
  - Use high-confidence test predictions as new training data
  - Help the model adapt to the test-domain distribution
- **Risk:**
  - Wrong pseudo labels may reinforce model mistakes
  - Confidence and margin filtering are used to reduce noise
- **Pseudo-label Selection:**
  - Top-1 prediction is not new_individual
  - Top-1 confidence is high enough
  - Top-1 score is clearly higher than top-2 score
- **Two Training Strategies:**
  - From-scratch pseudo training
  - Resume pseudo fine-tuning

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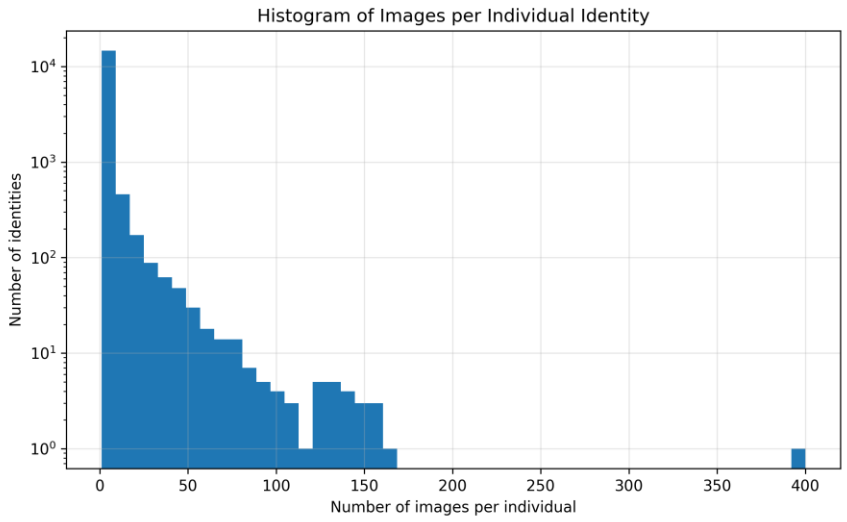
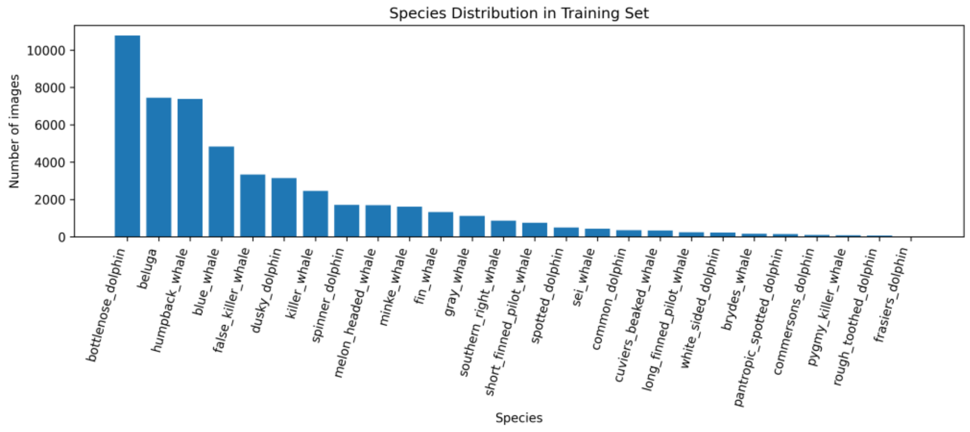
Dataset and Evaluation Metrics

- **Dataset:**
  - Image: 51,033/27956
  - Class: 15,587 IDs
  - Species: 26 (After label correction)
- **Evaluation Metric:**
  - Mean Average Precision @ 5 (MAP@5)

A	B	C	D	E	
A	A	A	A	A	→ 1.0
B	D	E	A	X	→ 0.25

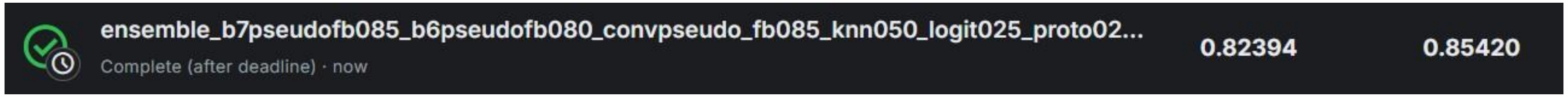




Comparison with state-of-the-arts

- **The gold-medal solution still performs better**
- **Our result shows improvement, but not yet SOTA-level**

Method	Setting	Public	Private
EfficientNet-B7	single model pseudo	0.84567	0.81535
Final ensemble	B7 + B6 + ConvNeXt-B pseudo	0.85420	0.82394
Gold-medal solution	multi-fold + multi-model	--	0.84909



Ablation studies: Pseudo-label

- **Pseudo-label fine-tuning did not improve B7**
- **Training from scratch with pseudo labels achieved the best B7 result**
- **This suggests that pseudo labels are useful when the model learns them from the beginning**

Setting	Public	Private
B7 without pseudo-labels	0.83531	0.80449
B7 pseudo, resume from epoch 30	0.83331	0.80155
B7 pseudo, from scratch	0.84567	0.81535

weighted_fb080_bf010_none010_knnlogit_species_newratio0.200_th0.321877 (2).csv Complete (after deadline) · 2d ago	0.80449	0.83531
weighted_fb080_bf010_none010_knnlogit_species_newratio0.200_th0.433184.csv Complete (after deadline) · 8h ago	0.80155	0.83331
weighted_fb080_bf010_none010_knnlogit_species_newratio0.200_th0.443228.csv Complete (after deadline) · 38m ago	0.81535	0.84567

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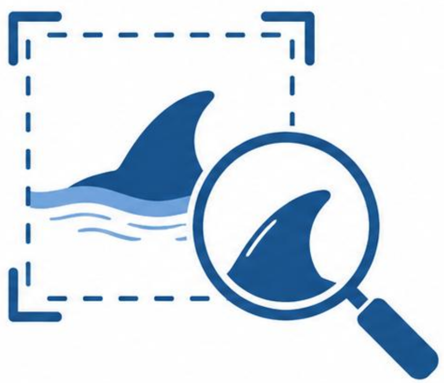
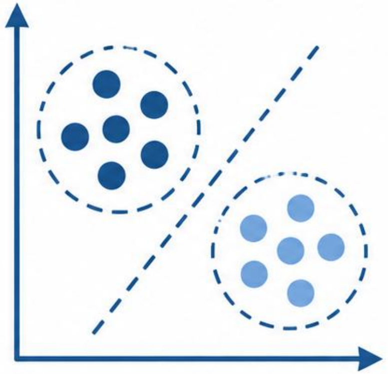
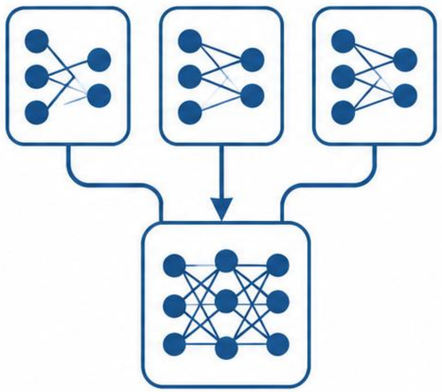
- **Our approach combines multi-view image inputs, task-specific metric learning, and model ensemble for fine-grained whale and dolphin identification.**







Data Preprocessing Matters

Useful identity cues such as scars, fin edges, body markings, and textures are sensitive to crop quality.

Task-specific Metric Learning

Sub-center ArcFace learns discriminative embeddings for fine-grained open-set retrieval.

Ensemble Improves Robustness

B7 + B6 + ConvNeXt-B improves over the strongest single B7 model.

Pseudo-label self-training helps

Pseudo labels improve B7 when training from scratch, but late fine-tuning may disturb the learned embedding space.

Team Member Contribution Table

Tasks	314511037 張周芳	313554063 劉霈琳	112550028 何柏翰
Literature survey	34%	33%	33%
Approach design	30%	55%	15%
Approach implementation (experiment)	40%	40%	20%
Report writing	75%	25%	0%
Slide making and oral presentation	50%	0%	50%